

FORM 000574/SPEC

CARIBBEAN EXAMINATIONS COUNCIL

ADVANCED PROFICIENCY EXAMINATION

SPECIMEN PAPER

MATHEMATICS

Unit 2 – Paper 02

$2\frac{1}{2}$ hours

This examination paper consists of three sections: Module 2.1, Module 2.2, and Module 2.3.

Each section consists of 2 questions.

The maximum mark for each section is 25.

The maximum mark for this examination is 150.

This examination consists of 6 pages.

INSTRUCTIONS TO CANDIDATES

DO NOT open this examination paper until instructed to do so.

Answer **ALL** questions from the **THREE** sections.

Unless otherwise stated in the question, all numerical answers **MUST** be given exactly **OR** to three significant figures as appropriate.

Allowed Examination Materials:

Mathematical formulae and tables (provided).

Electronic calculator.

• Ruler and compass.

SECTION 1 (MODULE 2.1)

1. Newton's Law of Cooling states that the rate at which the temperature of a hot object decreases is proportional to the difference between the temperature of the hot object and the constant temperature of its environment.

(i) Model this law using an appropriate differential equation connecting $\theta(t)$ and t , where $\theta(t)$ is the temperature of the hot object at time t , and α is the constant temperature of the environment. **[5 marks]**

(ii) Solve the differential equation in (i) above, and show that its solution is given by

$$\theta(t) = \alpha + \beta e^{-kt}$$

where β and k are constants, with $k > 0$. **[5 marks]**

(iii) A hot cup of coffee in a room had an initial temperature of 85°C . Five minutes later, its temperature was 55°C . The constant temperature of the room is 25°C .

(a) Using the above data, determine the values of β and k in the equation

$$\theta(t) = \alpha + \beta e^{-kt} \quad \textbf{[5 marks]}$$

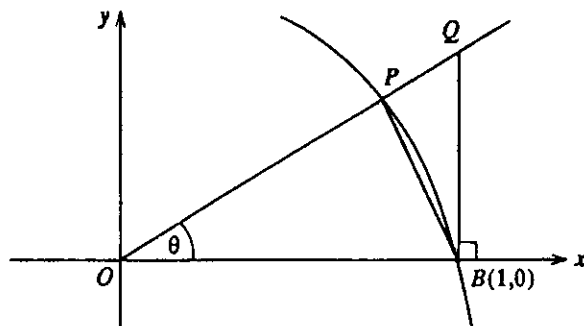
(b) Calculate the temperature of the cup of coffee after a further interval of five minutes. **[4 marks]**

(c) Sketch a graph showing the temperature of the cup of coffee as a function of time. **[6 marks]**

Total 25 marks

GO ON TO THE NEXT PAGE

2.



The origin O of the x - y plane is the centre of a circle of unit radius in the plane. B is the point with coordinates $(1, 0)$, and P is a point on the circle in the first quadrant, so that the arc BP subtends an angle θ radians at O . Q is the point on OP , produced, so that BQ is perpendicular to the x -axis.

- (i) Write down the co-ordinates of P in terms of θ . [3 marks]
- (ii) Write down the co-ordinates of Q in terms of θ . [3 marks]
- (iii) Show that area of triangle OPB is $\frac{1}{2} \sin \theta$. [3 marks]
- (iv) Calculate, in terms of θ , the areas of
 - (a) the sector OPB , and [1 mark]
 - (b) the triangle OQB [3 marks]
- (v) By considering the areas of triangle OPB , sector OPB and triangle OQB , show that

$$\cos \theta < \frac{\sin \theta}{\theta} < 1$$
 [4 marks]
- (vi) Hence, show that $\frac{\sin \theta}{\theta} \rightarrow 1$ as $\theta \rightarrow 0$. [3 marks]
- (vii) Hence, show from first principles that the derivative at x of the sine function is $\cos x$. [5 marks]

Total 25 marks

SECTION 2 (MODULE 2.2)

3. (a) Prove, by mathematical induction, that

$$\sum_{r=1}^n r^2 = \frac{1}{6}n(n+1)(2n+1). \quad [7 \text{ marks}]$$

Hence, or otherwise, calculate in terms of n , the value of

$$\sum_{r=1}^n (3r-1)^2. \quad [5 \text{ marks}]$$

- (b) Consider the geometric series with n^{th} term given by ar^{n-1} , where

$$r \neq 1 \text{ and } a \neq 0.$$

- (i) Obtain a formula for the value of the sum, S_n , of the first n terms of the series. [5 marks]
- (ii) Explain, in your own words and in terms of the behaviour of S_n , under what conditions the geometric series converges. [3 marks]
- (iii) If $r = \frac{4x}{3+x^2}$, find all values of x for which the geometric series will converge. [5 marks]

Total 25 marks

4. The function g is given by

$$g(x) = x^5 + x - 1.$$

Show that:

- (i) g is everywhere strictly increasing; [3 marks]
- (ii) the equation $x^5 + x - 1 = 0$ has a real root α in the interval $[0, 1]$; [4 marks]
- (iii) α is the only real root of the equation. [4 marks]
- (iv) Choose $x_1 = 1$ as a first approximation to α .
Sketch a diagram illustrating how the Newton-Raphson method leads to a second approximation x_2 to α . [5 marks]
- (v) Show that, if x_n is the n^{th} approximation to α , then

$$x_{n+1} = \frac{4x_n^5 + 1}{5x_n^4 + 1} \quad [5 \text{ marks}]$$

- (vi) Use the Newton-Raphson method to obtain an approximation to α , correct to three significant figures. [4 marks]

Total 25 marks

GO ON TO THE NEXT PAGE

SECTION 3 (MODULE 2.3)

5. (a) The following data, compiled in the categories shown, indicate the acceptance of the product *yam-pudding* by a group of persons:

PRODUCT ACCEPTANCE				
SEX	EXCELLENT	GOOD	FAIR	POOR
MALE	140	450	300	50
FEMALE	100	350	450	150

If a person is randomly selected from the group, what is the probability that the person -

- (i) considers the product to be "good"? [3 marks]
 - (ii) is female, and considers the product to be "fair"? [3 marks]
 - (iii) is male, or thinks the product to be "fair"? [5 marks]
 - (iv) will rate the product "excellent", given that the person selected is female; [5 marks]
- (b) A small box contains ten blue balls and four white balls. Three balls are drawn, one after the other. Use tree diagrams to find the probability of drawing one blue ball and two white balls, if -
- (i) the first two balls drawn are replaced; [5 marks]
 - (ii) the first two balls drawn are not replaced [4 marks]

Total 25 marks

6. (a) A teacher asked her students to draw a rectangle of length x cm and width y cm so that
- the length is at most 8 cm;
the width is at most 6 cm;
the perimeter is at least 20 cm.
- (i) For each of these three conditions, write down an inequality involving x or y or both, which models the given information. [6 marks]
- (ii) A student wrote down five inequalities to represent the situation completely. Write down the two other inequalities which are appropriate. [2 marks]
- (iii) Draw, on graph paper, each of the boundaries in the x - y plane of the five regions in which the inequalities defined in (i) and (ii) above are satisfied. Shade the region in which the points (x, y) satisfy the five inequalities are satisfied. [6 marks]
- (iv) State the dimensions of four rectangles that satisfy the five conditions. [3 marks]
- (b) At time t , the volume of a spherical drop of radius r is decreasing at a rate which is k times its surface area.
- (i) Write down an equation for the time rate of change of r . [5 marks]
- (ii) Show that, if the radius is initially a , then the drop will vanish after a time a/k . [3 marks]

Total 25 marks

END OF PAPER